perm. sym language

$$\Sigma = EE^T$$

"count fluctuation hyperplane"

$$p(w_1, w_2 \dots w_L) \propto e^{-\mathcal{E}[N(w_1, w_2 \dots w_L)]}$$

Notations.

Bag of words.

$t_\mu \to$ ctx pos $\quad$ i token $= 1, 2, .. N$

$\mu = 1, 2, \ldots L$.

$$p(t_1 .. t_L) = e^{-G_{ij} n_i n_j}.$$

$$n_i = \left(\sum_{\mu=1}^{L} \delta_{t_\mu i} - p_i \right) \qquad p_i = \langle n_i \rangle \text{ as a RV.}$$

choose $\vec{v}$ such that D_{KL} from prod. dist and the true dist is minimised.

$$\hat{p}(t_1 .. t_L) = \prod_{\mu=1}^{L} \frac{e^{\|v_{t_\mu}\|^2}}{Z(v_\mu)}, \qquad Z(v) = \sum_{\tilde{v}} e^{v \cdot v_{\tilde{v}}}$$

emb. then softmax

$$\text{or } Z(v) = \sum v \cdot v_{\tilde{v}}$$

$$D_{KL}(\hat{p} \| p) = \sum_{t_1 .. t_L} e^{-G_{ij} n_i n_j} \log \left(\frac{e^{-G_{ij} n_i n_j}}{\prod_{\mu=1}^{L} \frac{|v_\mu|^2}{Z(v_\mu)}} \right)$$

$$= \sum_{t_1 \ldots t_L} e^{-G_{ij} n_i n_j} \sum_{\mu=1}^{L} \left(\log(Z(v_\mu)) - \log v_\mu \cdot v_\mu \right)$$

$$= \sum_{t_1 .. t_L} e^{-G_{ij} n_i n_j} \sum_{l=1}^{N} n_l \left(\log Z(v_l) - \log v_l \cdot v_l \right)$$

$p_l + n_l$

$$\frac{\partial D_{KL}}{\partial v_k} = \sum_{t_1 .. t_L} e^{-G_{ij} n_i n_j} \sum_{l=1} \bar{n}_l \frac{1}{Z(l)} \left[\delta_{kl} \left(\sum v_m \right) + v_l \right] - \frac{2 v_l}{v_l \cdot v_l} \cdot \delta_{kl}$$

$$\frac{\partial Z_l}{\partial v_k} = \frac{\partial}{\partial v_k} \sum_m v_l \cdot v_m = \left[\delta_{kl} \left(\sum v_m \right) + v_l \right]$$

$$\frac{\partial D_{KL}}{\partial v_k} = \cancel{\sum_{t_1 .. t_L} e^{-G_{ij} n_i n_j}} \left[\bar{n}_k \left(\frac{\sum v_m}{Z(v_k)} - \frac{2 v_k}{|v_k|^2} \right) + \sum_{l=1} \bar{n}_l v_l / Z(v_l) \right]$$

Now rep. mean freq.

$$\frac{\partial D_{KL}}{\partial v_k} \cdot v_j = n_k \left(\frac{\sum_m W_{mj}}{\sum_m W_{mk}} - \frac{2 W_{jk}}{W_{kk}} \right) + \sum_{l=1} n_l \frac{W_{lj}}{\sum_m W_{ml}}$$

Approx: $Z(v_k) = \sum_l v_k \cdot v_l \approx v_k^2 = W_{kk}$.

$$\bar{n}_k \cdot \left(- \frac{2 v_k}{v_k \cdot v_k} \right) + \sum_l \bar{n}_l \cdot \frac{v_l}{v_l \cdot v_l} = 0.$$

Define $v_k / v_k \cdot v_k = w_k$, $\quad -2 \bar{n}_k w_k + \sum_l \bar{n}_l w_l = 0 \quad \forall k.$

$$k=1 \quad \begin{pmatrix} -n_1 & n_2 & n_3 & \cdots & n_N \end{pmatrix} \begin{pmatrix} - w_1 - \\ - w_2 - \\ \vdots \\ - w_n - \end{pmatrix} = 0.$$

$$k=2 \quad \begin{pmatrix} n_1 & -n_2 & n_3 & \cdots & n_N \end{pmatrix}$$

Try w/ a softmax model

$$D_{KL}(\hat{p}\|p) = \sum_{t_1..t_L} e^{-G_{ij} n_i n_j} \log\left(\frac{e^{-G_{ij} n_i n_j}}{\prod_{\mu=1}^{L} \frac{e^{v_\mu \cdot v_\mu}}{Z(v_\mu)}} \right)$$

$$D_{KL} = \sum_{t_1..t_L} e^{-G_{ij} n_i n_j} \left[-\sum_{\mu=1}^{L} \log e^{v_\mu \cdot v_\mu} - \log Z(v_\mu) \right]$$

$$= \sum_{t_1..t_L} e^{-G_{ij} n_i n_j} \sum_{i=1}^{N} (p_i + \underbrace{n_i}_{\text{vanishes when summed.}}) \left[-\log e^{v_i \cdot v_i} + \log Z(v_i) \right]$$

$$= \sum_{i=1}^{N} p_i \left(\log Z(v_i) - \log e^{v_i \cdot v_i} \right) \quad \rightarrow \quad D_{KL} = \sum_{i=1}^{N} -p_i \langle v_i \cdot v_i \rangle$$

$$= \sum_{i=1}^{N} p_i \left(\log \sum_{j=1}^{N} e^{v_i \cdot v_j} - \overbrace{v_i \cdot v_i}^{\text{norm}} \right)$$

minimize norm?

$$\frac{\partial D_{KL}}{\partial v_\ell} = \sum_{i=1}^{N} p_i \frac{1}{Z(v_i)} \underline{\frac{\partial Z(v_i)}{\partial v_\ell}} - 2 p_i \delta_{i\ell} v_i$$

$$\frac{\partial Z(v_i)}{\partial v_\ell} = \frac{\partial}{\partial v_\ell} \sum_{j=1}^{N} e^{v_i \cdot v_j} = \delta_{i\ell} \sum_{j=1}^{N} v_j e^{v_i \cdot v_j} + v_i \cdot e^{v_i \cdot v_\ell}$$

$$\frac{\partial D_{KL}}{\partial v_\ell} = \sum_{i=1}^{N} p_i \left[\frac{1}{Z(v_i)} \left(\delta_{i\ell} \sum_{j=1}^{N} v_j e^{v_i \cdot v_j} + v_i \cdot e^{v_i \cdot v_\ell} \right) \right] - 2 p_\ell v_\ell$$

$$= p_\ell \left[\frac{1}{Z(v_\ell)} \sum_{i=1}^{N} v_i e^{v_\ell \cdot v_i} \right] + \sum_{i=1}^{N} p_i \cdot \frac{e^{v_i \cdot v_\ell} v_i}{Z(v_i)} - 2 p_\ell v_\ell \overset{!}{=} 0$$

$$\frac{1}{Z} \sum E e^{-\beta E}$$

$$e_i \cdot e_j = \frac{\Sigma_{ij}}{L p_i p_j}$$

$$\left(\sum N_i e_i\right)^2 = \sum N_i N_j \left(\frac{\Sigma_{ij}}{L p_i p_j} \right)$$

$$= \sum (\sqrt{L} x_i + L p_i)(\sqrt{L} x_j + L p_j) \cdot \frac{\Sigma_{ij}}{L p_i p_j}$$

$$= \sum \left(\frac{\Sigma_{ij}}{p_i p_j} \right) x_i x_j + \sum L^{3/2} p_i x_j / p_i p_j \; \Sigma_{ij} + \text{const.}$$

$$\sum L^{3/2} \frac{\Sigma_{ij}}{p_j} x_j$$

22/03/26.

$$e^{-G_{ij}}$$

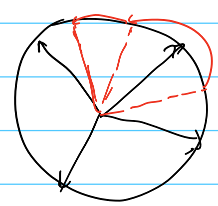
Sanjeev. Arora:

- Rand. Walk on a sphere.
- each time step: overlap with fixed "emb" as prob.
- Rand Walk generates seq.

embedding sampled from $v = s \cdot \hat{v}$ Gaussian. spherical uniform.

link to coocurence: $PMI = v_i \cdot v_j$ = overlap between embeddings.

underlying generative process is an RW.



with fixed stepsize d, closer words more likely to co-occur.

in ctx window of 2

Thm:

$$\log p(w, w') = \frac{\|v_w + w_{w'}\|^2}{2d} - 2\log Z \pm \epsilon.$$

$$\log p(w) = \frac{\|v_w\|^2}{2d} - \log Z + \epsilon$$

taken to be a constant